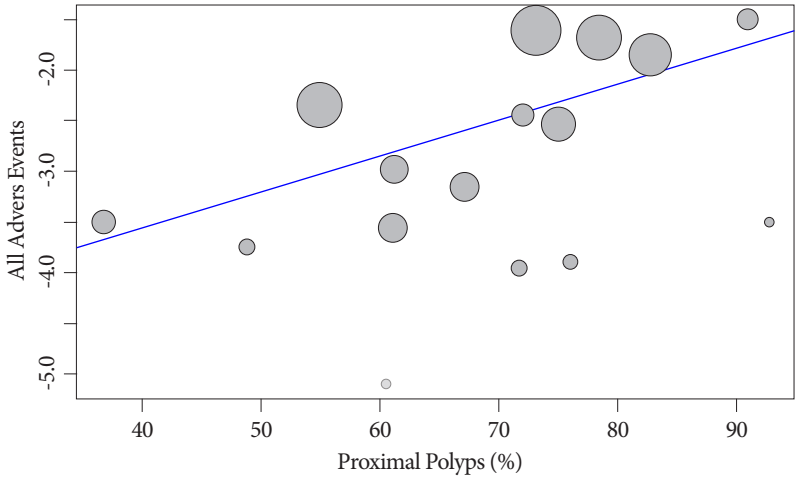
SUPPLEMENTARY MATERIALS



Supplementary Fig. 1. Scatterplot showing the relationship of proximal lesions and adverse events on meta-regression.

Supplementary Table 1. Data on Histopathology of Polyps Removed Using Underwater Endoscopic Mucosal Resection

Study	Number of Polyps Sessile ≥10 mm	Hyperplastic/ Infammatry	Tubular adenoma	Tubulovillous adenoma	Villous adenoma	Sessile serrate adenoma	High grade dysplasis	Adenocarcinoma	Comments
Amato et al. (2016)[39]	25	0	9	8	1	7	0	0	
Siau et al. (2018) [47]	102	0	45	33	0	24	0	0	102
Schenck et al. (2017)[46]	73	5	31	17	0	14	5	1	73
Curcio et al. (2015)[43]	81	7	21	4	0	4	34	11	
Uedo et al. (2015)[48]	11	1	6	3	0	1	0	0	
Binmoeller et al. (2015)[18]	53	0	26	14	2	10	0	1	53
Binmoeller et al. (2012)[16]	62	0	22	19	4	11	5	1	62
Wang et al. (2013)[20]	43	3	29	5	0	3	3	0	
Yamashina et al. (2019)[49]	108	0	-	-	-	17	0	18	total 70 adenoma, 3 others
Chien et al. (2019)[42]	171	0	-	-	-	20	63	11	adenoma 77
Chaves et al. (2018)[41]	16	0	0	0	0	16	0	0	

Supplementary Table 2A. Study Quality Assessment Based on Newcastle-Ottawa Scale

Author	Representativeness of the average adult in community	Cohort size	Information on clinical outcomes	Outcome not present at start	factors comparable between the groups	Adequate clinical assessment	Follow up time	Adequacy of follow-up	MAX=8	HIGH>6, MEDIUM 4 to 6, LOW ≤3
Amato et al. (2016)[39]	0	0.5	1	1	1	1	1	0	5.5	Medium
Siau et al. (2018) [47]	0.5	1	1	1	1	1	1	0.5	7	High
Barclay et al. (2020)[17]	0	1	1	1	1	1	1	0.5	6.5	High
Sandhu et al. (2018)[45]	0	1	1	1	1	1	1	0.5	6.5	High
Schenck et al. (2017)[46]	0	1	1	1	1	1	1	0.5	6.5	High
Curcio et al. (2015)[43]	0	1	1	1	1	1	1	1	7	High
Uedo et al. (2015)[48]	0	0	1	1	1	1	0	0	4	Medium
Binmoeller et al. (2015)[18]	0	1	1	1	1	1	1	0.5	6.5	High
Binmoeller et al. (2012)[16]	0	1	1	1	1	1	1	0.5	6.5	High
Cadoni et al. (2018)[40]	0.5	1	1	1	1	1	1	0.5	7	High
Wang et al. (2013)[20]	0	0.5	1	1	1	1	0	0	4.5	Medium
Rodríguez Sánchez et al. (2019)[44]	0.5	1	1	1	1	1	1	0.5	7	High
Chien et al. (2019)[42]	0.5	1	1	1	1	1	0	0	5.5	Medium
Chaves et al. (2018)[41]	0	0	1	1	1	1	0	0	4	Medium

Supplementary Table 2B. Showing Study Quality Assessment Based on Jadad Score for Randomized Controlled Trials

Study	Randomization	Randomization generation	Double blinding	Allocation concealment	Withdrawals or dropouts	Jadad score (MAX=5)	Quality
Yamashina et al. (2019)[49]	1	1	0	0	1	3	Good
Yen et al. (2019)[50]	1	1	0	0	1	3	Good

Supplementary Table 3. Data on Assessed Outcomes for Nonpedunculated Polyps ≥20 mm

Author, year	Number of polyps (*n*)	Enbloc resection (*n*)	Incomplete resection (*n*)	R0 resection (*n*)	Total Complications (*n*)	Intra-procedural bleeding (*n*)	Delayed Bleeding (*n*)	Perforation (*n*)	Post-polypectomy syndrome (*n*)	Recurrence per patient
Amato et al. (2016)[39]	18	12	0	12	NR		0	0	0	NR
Barclay et al. (2020)[17]	264	74	3	NR	44	39	4	0	1	10/174
Schenck et al, 2017	40	3	NR	NR	2	0	2	0	0	3/40
Uedo et al. (2015)[48]	7	5	NR	5	NR	NR	0	0	0	NR
Binmoeller et al. (2015)[18]	53	29	0	23	1	0	1	0	0	2/50
Cadoni et al. (2018)[40]	18	7	NR	7	5	5	0	0	0	NR
Rodríguez Sánchez et al. (2019)[44]	12	5	0	NR	NR	NR	NR	0	0	NR
Chien et al. (2019)[42]	42	29	NR	NR	NR	2	NR	NR	0	NR
Chaves et al. (2018)[41]	10	3	NR	4	0	0	0	0	0	NR
Yen et al. (2019)[50]	16	4	1	-	3	3	0	0	0	0/12

NR: not reported.

Supplementary Table 4. Meta-Regression Results

	All complications		Intra-procedural bleeding		Recurrence rate	
	Coefficient with *p*-value*	R^2	Coefficient with *p*-value	R^2	Coefficient with *p*-value	R^2
Percentage of Proximal lesions	0.03 (0.005 - 0.064), ***p***=0.02	34.5%,	0.04 (0.006 - 0.072), *p*-0.01	34.3%	-0.027 (-0.056 - 0.009), *p*=0.06	61.6%
Study type	0.67 (-0.29 - 1.64), *p*=0.17,	0%	0.45 (-0.71 - 1.62), *p*=0.44	0%	0.35 (-0.99 - 1.71), *p*=0.6	0%
Mean polyp size	0.009 (-0.042, 0.061), *p*=0.73,	0%	0.004 (-0.058 - 0.067), *p*=0.9	0%	-0.03 (-0.105 - 0.038), *p*=0.35	19.6%
Mean Age	0.007 (-0.12 - 0.14), p=0.91	0%	0.033 (-0.11 - 0.178), p=0.64	0%	0.13 (-0.11 - 0.37), p=0.25	19.4%

*Values are reported with regression coefficient and 95% confidence interval. R^2 is the % of heterogeneity accounted for outcome of interest after adjusting for the variable. Bold indicates significant value

APPPENDICES

Appendix 1. Literature search strategy.
Appendix 2. Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) checklist.
Appendix 3. Meta-analysis of Observational Studies in Epidemiology (MOOSE) checklist.